Body Composition and Weight Changes during a 15-Week Marathon Training Program

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Abstract

With the increasing rates of obesity worldwide, exercise and nutritional interventions have been at the forefront in attempts to produce positive health changes. The general public often views the effectiveness of an exercise program solely on changes in their weight. This research, demonstrates that weight loss may not occur without months of intervention and that weight loss may be a poor assessment to quantify immediate positive results. This study also gives evidence that the public needs a practical way to access body composition changes through intervention programs designed to lower body weight. Data was collected from a Marathon Training course offered at the University of Minnesota-TC through three consecutive years (2009-2011). Body composition and body weight were compared from a pre/post 15-wk structured marathon training plan. Results provided evidence that significant changes in body composition occurred for both male and females, however not body weight. This phenomenon was more prevalent in females, who actually gained weight, which is consistent with previous literature looking at the dichotomy of gender differences (Bjorntorp, 1980).

This data further illustrates the importance of minimizing the use of body weight as a solitary measurement of immediate changes in health gains through exercise.

Methods

Subjects: Eighty-five students (30 males, 55 females) were recruited from a marathon training course offered by the Physical Activity Program at the University of Minnesota (PE 1262). They included VO_2max testing, body composition testing and other standard measurements for calculation purposes, such as height, weight, etc. The researchers measured each participant for weight, height, and % BF prior to the training and again within two weeks prior to the marathon after 15 weeks of marathon training. Body composition (i.e., percentage of body fat) was assessed by BodPod measurements. BMI was calculated using the equation weight (kg) / height^2(m). All of these measurements were taken at the onset of training and at the post-test.

Data were analyzed using SPSS (v19.0). Means, 95% confidence intervals, and results based on a 0.05 significance level are reported. A Levene's test of homogeneity indicated that the data exhibited constant variance, meeting the assumptions for parametric tests. Independent *t*-tests were used to compare males and females on three dependent variables: (a) change of weight (kg), (b) change in BMI, and (c) change in percentage of body fat after 15 weeks of marathon training. Differences between pre- and post-test measurements of weight, BMI, and percentage of body fat were assessed using dependent *t*-tests for paired samples.

Descriptive Data

Age of Participants

	Mean Age (y)		Age Range (y)	
Year	Female	Male	Female	Male
2009	20.66	21.00	18-24	19-22
2010	21.06	21.80	18-28	18-26
2011	20.89	22.43	18-26	19-37
Overall	20.89	21.74	18-28	18-37

Women Pre- and Post-Test Values

	Mean Weight (kg)		Mean BMI		Mean % Body Fat	
Year	Pre	Post	Pre	Post	Pre	Post
2009	60.75	60.73	22.28	22.27	23.57	21.99
2010	64.82	64.65	23.41	23.35	24.47	23.21
2011	65.92	66.23	23.22	23.37	28.00	27.29
Overall	63.83	63.87	22.97	23.00	25.36	24.16

Men Pre- and Post-Test Values

	Mean Weight (kg)		Mean BMI		Mean % Body Fat	
Year	Pre	Post	Pre	Post	Pre	Post
2009	80.19	78.09	24.45	23.61	13.72	12.18
2010	73.86	72.83	22.55	22.26	10.36	08.19
2011	75.17	75.56	22.78	23.01	14.60	14.06
Overall	76.41	75.50	23.26	22.96	12.89	11.46

Results

In this study, weights, the body mass index, and percentages of body fat were measured on 96 study volunteers from a three-credit marathon training course offered by the physical activity program at the University of Minnesota (PE 1262) in 2009, 2010, and 2011. Data from nine volunteers were removed from the study due to recording errors and two additional cases were removed due to outliers. Thus the present results are drawn from 85 participants. Data were evaluated with Levene's tests for constant variance, and were found to meet the assumptions of the two-tailed independent and paired samples *t*-tests performed for this study. Average changes between male and female participants on each of the measures (weight, body mass index, and percentages of body fat) are presented in Table 4 and summarized in Figure 1.

Summary Data

Weight Changes

Gender	Sample size (*n*)	Average Pre-Training Weight (kg)	Average Post-training Weight (kg)	Average Change in Weight (kg)	Significance of difference in weight and effect size
Female	55	64.70 (*SE* = 1.19)	64.91 (*SE* = 1.14)	0.21 (*SE* =.22)	p = .346, r = .13
Male	30	76.24 (*SE* = 1.68)	75.50 (*SE* = 1.57)	-0.74 (*SE* =.39)	p = .075, r = .32
Total	85	67.88 (*SE* = 1.15)	68.25 (*SE* = 1.08)	0.13 (*SE* = .202)	p = .525, r = .07

Table 1. Average changes in **body weight** after marathon training for males and females and statistical significance of these changes.

BMI Changes

Gender	Sample size (n)	Average Pre-Training BMI	Average Post-training BMI	Average Change in BMI	Significance of difference in BMI and effect size
Female	55	23.11 (SE = .36)	23.20 (SE =.35)	.09 (SE =.08)	p = .270, r = .15
Male	30	22.99 (SE = .42)	23.16 (SE =.44)	-.17 (SE =.13)	p = .191, r = .16
Total	85	23.127 (SE = .27)	23.123 (SE =.27)	-.004 (SE = .07)	p = .952, r = .01

Table 2. Average changes in **BMI** after marathon training for males and females and statistical significance of these changes.

% Body Fat Changes

Gender	Sample size (*n*)	Average Pre-Training % body fat	Average Post-training % body fat	Average Change in % body fat	Significance of difference in % body fat and effect size
Female	55	26.65 (*SE* = .85)	25.61 (*SE* =.79)	-1.04 (*SE* =.35)	p = .004, r = .38
Male	30	13.19 (*SE* = 1.21)	11.93 (*SE* =1.11)	-1.26 (*SE* =.46)	p = .010, r = .45
Total	85	21.90 (*SE* = .98)	20.78 (*SE* =.96)	-1.12 (*SE* =.27)	p < .001, r = .42

Table 3. Average changes in **percentages of body fat** after marathon training for males and females and statistical significance of these changes.

Literature Cited

Bjorntorp, P. A., (1980) Sex differences in the regulation of energy balance with exercise. *The American Journal of Clinical Nutrition, 49,* 958-61.

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2009-2011 Marathon Class Participants

McNair Staff

Results Continued

Summary Data

Differences in Men and Women

	Female	Male	Average difference between female and male participants	Statistical significance of difference between female and male participants
Average change in weight	.21 (SE =.22)	-.74 (SE =.40)	.95 kg (*SE* =.41)	p = 0.027
Average change in BMI	.09 (*SE* =.08)	-.18 (*SE* =.12)	.258 (*SE* =.14)	p = 0.073
Average change in % body fat	-1.04 (*SE* =.35)	-1.24 (*SE* = .44)	.206 (*SE* =.57)	p = 0.718

Table 4. Average changes in weight, BMI, and percentage of body fat after marathon training for female and male participants and differences in weight, BMI, and percentage of body fat between female and male participants.

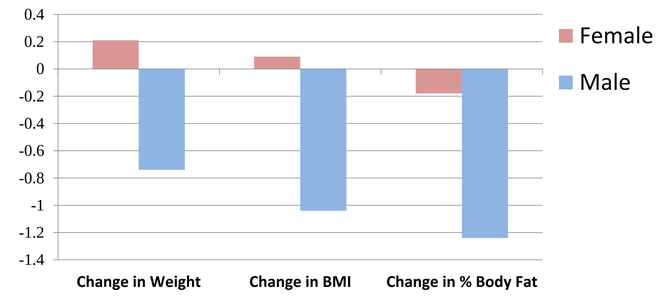


Figure 1. Changes in weight, BMI, and percentage of body fat after marathon training for female and male participants.

Summary

These findings demonstrate similar patterns of change over the course of marathon training for men and women. Most male participants lost weight, body mass, and body fat. While most women gained weight and body mass, these increases were almost so small as to be negligible (.21 kg increase in weight and a .09 increase in BMI, respectively). The average decrease in body fat was much greater for all participants, with 55% of the female participants and 67% of male participants recording a loss in body fat. These results suggest that marathon training is effective in reducing body fat for most participants, both male and female, but that a very small increase in weight and body mass may be expected for females involved in marathon running training programs.

Based on evidence from this research, body weight is not an appropriate measure in the first 15 weeks of an activity intervention for health gains. In sedentary to moderately active college-age students, an aggressive endurance training program did not illicit weight loss that was significant in 85 students. However, percent of body fat did decrease which was statistically significant in both men and women. The gains in muscle mass and the associated weight, as well as potentially increased water weight with increased mitochondrial stores that happens in endurance training, initially offsets any weight loss, particularly in women.

Weight loss is one of the major motivators for continued effort in a weight loss program. It is important to inform populations that are attempting to lose weight, that while body composition is improving in a positive direction with fat loss, there are few practical measures available to assess these gains in health.